

Soham Hariprasad Katkar

Software Engineer

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Professional Summary

Software Engineer with nearly **4 years of experience** across US and global environments, specializing in Python-based backend development, cloud-native systems, and CI/CD automation. Proven background in designing RESTful APIs, supporting containerized applications, and maintaining production reliability for enterprise platforms. Experienced in AWS and Azure environments, collaborative Agile teams, and data-driven problem solving, with a strong focus on scalable architecture, automation, and operational efficiency.

Technical Skills

- Programming & Scripting:** Python, Java, C++, JavaScript (ES6+), SQL, PL/SQL, Bash, Unix/Linux Shell Scripting
- Backend & Web Technologies:** FastAPI, Flask, Spring Boot, Node.js, Express.js, RESTful APIs, Microservices Architecture
- Frontend Technologies:** React.js, HTML5, CSS3, Responsive UI Development
- Cloud & Infrastructure:** AWS, Azure, Docker, Kubernetes, Terraform, AWS CloudFormation, Ansible
- CI/CD & Automation:** Jenkins, GitHub Actions, CI/CD Pipelines, Apache Airflow
- Data & AI Technologies:** Pandas, NumPy, Neo4j, LangChain, Ollama, ChromaDB, DVC
- Monitoring & DevOps Tools:** Splunk, Prometheus, GitHub, Jira
- AI & Developer Tools:** OpenAI APIs, Gemini, Groq, Cursor

Professional Experience

Software Engineer | HCLTech Jul 2025 – Present | United States

- Developed Python-based backend services and automation utilities supporting enterprise applications, focusing on secure integrations, maintainable service design, and reliable communication across distributed systems.
- Designed RESTful APIs using Python frameworks and Java services, implementing authentication, request validation, logging, and error handling aligned with enterprise engineering standards.
- Built and maintained CI/CD pipelines using Jenkins and GitHub Actions, automating builds, testing, and deployments for containerized applications on Docker and Kubernetes.
- Improved deployment stability by refining release validation processes and monitoring workflows, reducing environment-related issues and improving coordination across engineering and operations teams.
- Enhanced system observability using Splunk and Prometheus, enabling faster incident diagnosis and reducing mean time to resolution for production support activities.

Software Engineer | HSBC Technology Aug 2021 – Dec 2023 | Pune, India

- Developed backend components using Python, Java, and C++ for global payment systems, supporting transaction processing, service orchestration, and integration with multiple enterprise platforms.
- Created Python automation scripts for CI/CD workflows, server provisioning, and deployment validation, standardizing processes across development, testing, and production environments.
- Expanded backend testing frameworks using JUnit and Python-based utilities, strengthening validation logic for critical payment workflows and improving overall code reliability.
- Improved system performance by optimizing SQL queries and Python data-processing routines, reducing execution latency and supporting increased transaction throughput.
- Resolved over two hundred production issues through structured debugging, root cause analysis, and peer code reviews, contributing to measurable improvements in platform stability.

Data Analyst Intern | Incentius Solutions Ltd. Feb 2021 – Jul 2021 | Pune, India

- Built Python-based data processing workflows to clean, validate, and transform large sales datasets sourced from multiple regional systems.
- Developed SQL queries and automated ETL pipelines to aggregate, validate, and analyze sales performance metrics for analytics teams.
- Designed interactive dashboards using Power BI and Excel to visualize trends, regional performance, and operational metrics.
- Reduced reporting turnaround time by eliminating redundant manual steps and optimizing transformation logic within recurring workflows.
- Increased stakeholder adoption by delivering consistent, well-documented dashboards aligned with leadership reporting requirements.

Academic & Technical Projects

PeerAnalytica – AI Academic Advisory Platform

- Developed an AI-driven academic advisory system that analyzes student behavioral and academic data using graph-based relationships to generate personalized study recommendations through backend services and interactive dashboards.
- Improved academic guidance accuracy and response consistency by validating recommendations across multiple student profiles while ensuring data privacy and reliable system performance.
- Utilized Python, Flask, Neo4j, React.js, RESTful APIs, and local LLM integration using Ollama for backend processing, analytics generation, and responsive frontend visualization.

RAG Assistant for Credit Card Policies

- Built a retrieval-augmented question answering system to provide accurate, document-grounded responses for credit card policy queries using structured document ingestion and semantic search workflows.
- Increased response relevance and reduced hallucinated outputs by optimizing document chunking, retrieval parameters, and prompt workflows across multiple policy documents.
- Applied Python, LangChain, ChromaDB, Ollama, and Streamlit to implement vector storage, retrieval orchestration, model inference, and an interactive user-facing application.

Chicken Disease Classification using Deep Learning and MLOps

- Developed an end-to-end machine learning application to classify poultry diseases from image data, covering model training, validation, deployment automation, and cloud-based inference workflows.
- Reduced deployment turnaround time and improved reproducibility by implementing automated CI/CD pipelines, containerized deployments, and structured experiment tracking across environments.
- Used Python, TensorFlow, Docker, GitHub Actions, DVC, AWS, and Azure to manage model training, versioning, continuous integration, and scalable cloud deployments.

Education

Master of Science, Information Systems	Dec 2025
University of Maryland, Baltimore County	GPA: 3.62 / 4.0
Bachelor of Engineering, Electronics and Telecommunications	Jun 2021
Savitribai Phule Pune University	GPA: 3.30 / 4.0

Certifications

- Google Fundamentals of Generative AI
- Databricks Generative AI Certification